Which Condition Fits? Multi-Label Diagnosis with Supporting Evidence from Symptom Description

Niyaaz Baines
University of Victoria
nbaines@uvic.ca

Eduardo Kallina de Paula
University of Victoria
ekallinadepaula@uvic.ca

Thiago Vasconcelos Pinheiro Lopes
University of Victoria
thiagovplopes@uvic.ca

Priyanshkumar Ghanshyambhai Patel
University of Victoria
priyanshkumarghansh1@uvic.ca

ABSTRACT

ADHD, anxiety, and depression share symptoms such as inattention, restlessness, and irritability, yet most text classifiers assign one condition per description. We propose CoRE (Co-occurring condition classification with Rationale Extraction), which returns the best-fitting condition for a self-reported symptom description, any other consistent conditions, and the phrases that support each. CoRE learns from Reddit posts with weak multi-label targets drawn from the source community and from self-reported diagnoses, masks condition names, models co-occurrence with an order-averaged classifier chain, and selects evidence by phrase occlusion. The seeded implementation reproduces its results from one quickstart script. Code: github.com/eduardokdp/Which-Condition-Fits-.

KEYWORDS

Mental health, multi-label classification, classifier chains, rationale extraction

1 INTRODUCTION

Most text-based methods reviewed in Milestone 2 separate one condition from healthy controls or assign one label per post. Multi-label work such as SMHD [1] and the comorbidity ensemble of Souza et al. [9] gives no evidence for its predictions, and rationale methods [3, 4] have not been applied to co-occurring conditions. CoRE combines existing techniques into a new pipeline with three parts: (i) weak multi-label supervision from self-reported diagnoses, with condition names masked; (ii) an order-averaged classifier chain in which the score of one condition depends on the others; and (iii) per-condition occlusion rationales that expose phrases shared by several conditions.

2 PROBLEM DEFINITION: INPUT AND OUTPUT

Let $D = \{d_1, \ldots, d_k\}$ be an ordered set of conditions ($k = 3$: ADHD, anxiety, depression) and S a set of self-reported descriptions. A segmentation function σ splits $s \in S$ at sentence ends and at clause boundaries where a comma or conjunction precedes a subject such as "I" or "my", giving phrases $\sigma(s) = \langle p_1, \ldots, p_n \rangle$, $n \geq 1$; $s \setminus p_j$ is s with p_j erased. Refining Milestone 1, CoRE computes

$$f(s) = \big((d^*, E^*),\ \{(d_i, E_i)\}_{i=1}^{m}\big), \quad 0 \leq m \leq k-1, \quad (1)$$

where $d^* \in D$ is the primary condition, $d_1, \ldots, d_m \in D \setminus \{d^*\}$ are distinct secondary conditions, and $E^*, E_i \subseteq \sigma(s)$ are their evidence sets. The secondary set may be empty, and a phrase may belong to several evidence sets, because a symptom such as restlessness can support more than one condition. Figure 1 follows the running example of Milestone 1, whose description splits into the phrases p_1, p_2, and p_3, from input to output; Algorithm 1 gives the pseudo-code.

3 PROPOSED METHOD

Weak multi-label supervision. A training post s comes from a community $c(s)$ (r/ADHD, r/Anxiety, r/depression); c_d is the community of d. Its target is

$$y_d(s) = \mathbf{1}[c(s) = c_d] \ \vee \ \mathbf{1}[d \in R(s)], \quad (2)$$

where $R(s)$ holds the conditions named in first-person diagnosis statements ("I was diagnosed with", "I also have", "my..."), rejecting negated, hedged ("I wonder if I have"), and third-party ("my brother has") ones. Community labels alone are single-label, whereas self-reports are precise and are how CLPsych and SMHD label users [1, 2]. A mask μ then replaces condition names, but not symptom words such as "anxious", with one token, so the model must rely on symptom language.

Representation. The vector $\mathbf{x} = \phi(\mu(s))$ is a sublinear TF-IDF vector over word uni- and bigrams t formed between masked names, so the mask is never a feature:

$$x_t = \frac{(1 + \ln \mathrm{tf}_t)\, \mathrm{idf}_t}{Z}, \qquad \mathrm{idf}_t = \ln \frac{1+N}{1+\mathrm{df}_t} + 1, \quad (3)$$

with $\mathrm{tf}_t \geq 1$ the count of t in $\mu(s)$ ($x_t = 0$ if t is absent), df_t the number of the N training posts containing t, and Z the ℓ_2 norm. It keeps the pronoun "I" and contractions such as "can't", since first-person pronouns mark depression [8]. A linear representation is cheap to recompute, which occlusion needs.

Order-averaged classifier chain. Binary relevance scores each condition alone and ignores co-occurrence; a classifier chain [7] conditions each label on the previous ones but depends on an arbitrary order. For every order $\pi \in \mathrm{Perm}(D)$, link j computes

$$q^{\pi}_{\pi_j} = \mathrm{sig}\big(\mathbf{w}^{\top}_{\pi,j}[\mathbf{x};\ q^{\pi}_{\pi_1}, \ldots, q^{\pi}_{\pi_{j-1}}] + b_{\pi,j}\big), \quad (4)$$

and $P(d \mid s) = \frac{1}{k!} \sum_{\pi} q^{\pi}_d$ averages the $3! = 6$ chains. The order matters: for the running example, a single chain's ADHD score ranges from 0.67 to 0.86 across orders (Figure 1), and the mean removes this dependence. Each link is an ℓ_2-regularised, class-weighted logistic regression ($C = 1$). In training, the predecessor

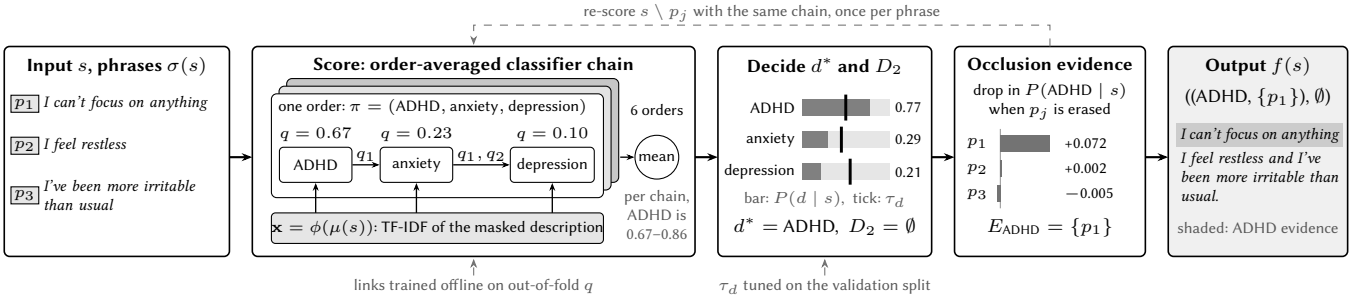


Figure 1: CoRE on the running example, with the quickstart model's values. The chain box unrolls one of the six label orders; occlusion re-scores the description once per erased phrase, for every predicted condition.

Algorithm 1 CoRE training and inference

Input: posts $\mathcal{S}$ with communities; conditions D; description s
Output: $f(s) = ((d^*, E_{d^*}), \{(d, E_d) : d \in D_2\})$

for $t \in \mathcal{S}$ **do** $\mathbf{y}(t) \leftarrow$ Eq. (2); $t \leftarrow \mu(t)$
split $\mathcal{S}$ 70/15/15 (stratified, seed 42); $X \leftarrow \phi(\mathcal{S}_{tr})$
for $\pi \in \mathrm{Perm}(D)$ **do** $Q \leftarrow \mathbf{0}$
  for $j = 1$ **to** k **do** $X_j \leftarrow [X;\, Q_{\pi_1 .. \pi_{j-1}}]$
    $Q_{\pi_j} \leftarrow \textsc{OutOfFold}(X_j, \mathbf{y}_{\pi_j})$; $h_{\pi,j} \leftarrow \textsc{LogReg}(X_j, \mathbf{y}_{\pi_j})$
$\tau_d \leftarrow \arg\max_\tau \mathrm{F1}_d(\tau)$ on $\mathcal{S}_{va}$, for each $d \in D$
$\langle p_1, \ldots, p_n \rangle \leftarrow \sigma(s)$; $\mathbf{P} \leftarrow \textsc{Score}(s)$
$d^* \leftarrow \arg\max_d P_d$; $D_2 \leftarrow \{d \neq d^* : P_d \geq \tau_d\}$
for $d \in \{d^*\} \cup D_2$ **do**
  $\Delta_d(p_j) \leftarrow P_d - \textsc{Score}(s \setminus p_j)_d$ for $j = 1, \ldots, n$
  $E_d \leftarrow \mathrm{top}_{\lceil n/2 \rceil}\{p_j : \Delta_d(p_j) > \delta\}$

$\textsc{Score}(t)$: mean over π of the chain outputs (Eq. 4) on $\phi(\mu(t))$

inputs are 5-fold out-of-fold probabilities, so each link sees the same kind of input as at inference.

Decision rule. The proposal requires a primary condition for every description, so $d^* = \arg\max_d P(d \mid s)$, and $D_2 = \{d \neq d^* : P(d \mid s) \geq \tau_d\}$ are the secondary conditions. Each threshold τ_d is the value on a 0.05-step grid that maximises the F1 of d on the validation split, which corrects for different base rates.

Occlusion rationales. The contribution of p_j to d is $\Delta_d(p_j) = P(d \mid s) - P(d \mid s \setminus p_j)$, and E_d keeps the top $\lceil n/2 \rceil$ phrases with $\Delta_d(p_j) > \delta = 0.01$. Erasure [5] needs no rationale labels, works with any scorer, and targets ERASER comprehensiveness [3]. A phrase in two evidence sets is reported as a shared symptom.

4 IMPLEMENTATION

The repository follows the FAIR principles [10] and the course template: a driver, `src/main.py`, calls `dataprep` (Eq. 2, masking, splits), `core` (one module per stage of Section 3), and `eval` (metrics and baselines), which share `cmn` (configuration, seeding, I/O). The full run uses the 2019 r/ADHD, r/Anxiety, and r/depression files of the public-domain Reddit Mental Health Dataset [6], with up to 5,000 posts per community after removing deleted posts, bots, duplicates, and all but one post per author, which prevents author leakage across splits.

Quickstart and reproducibility. `quickstart.sh` installs the pinned requirements, runs 18 unit and reproducibility tests, and runs the pipeline in about a minute on 1,786 public Reddit posts bundled with the code (350 per community used). Because r/ADHD and the other two communities come from different public collections, these scores are indicative only. Seeding (42) and a SHA-256 check of the predictions make runs repeatable. On 158 test posts, CoRE reaches a micro-F1 of 0.65 (95% bootstrap CI 0.60–0.70), within noise of both baselines, so whether the chain's extra cost (18 logistic regressions instead of 3) pays off is left to Milestone 4; deleting its evidence lowers the predicted probability by 0.12, against 0.02 for as many random phrases. For the running example it returns ADHD with evidence p_1 (Figure 1), the primary condition of the Milestone 1 target.

Contributions. Niyaaz: formal definition, abstract, introduction; Eduardo: method, references; Thiago: pseudo-code, implementation; Priyansh: Figure 1, integration. **Use of AI** (Level D): parts of the code and text were drafted with Claude (Anthropic), then reviewed, run, and revised by the authors.

REFERENCES

[1] Arman Cohan, Bart Desmet, Andrew Yates, et al. 2018. SMHD: A Large-Scale Resource for Exploring Online Language Usage for Multiple Mental Health Conditions. In *Proc. COLING*. 1485–1497.
[2] Glen Coppersmith, Mark Dredze, Craig Harman, et al. 2015. CLPsych 2015 Shared Task: Depression and PTSD on Twitter. In *Proc. CLPsych Workshop*. 31–39.
[3] Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, et al. 2020. ERASER: A Benchmark to Evaluate Rationalized NLP Models. In *Proc. ACL*. 4443–4458.
[4] Tao Lei, Regina Barzilay, and Tommi Jaakkola. 2016. Rationalizing Neural Predictions. In *Proc. EMNLP*. 107–117.
[5] Jiwei Li, Will Monroe, and Dan Jurafsky. 2016. Understanding Neural Networks through Representation Erasure. arXiv:1612.08220.
[6] Daniel M. Low, Laurie Rumker, Tanya Talkar, et al. 2020. Natural Language Processing Reveals Vulnerable Mental Health Support Groups and Heightened Health Anxiety on Reddit during COVID-19. *J. Medical Internet Research* 22, 10 (2020), e22635.
[7] Jesse Read, Bernhard Pfahringer, Geoff Holmes, and Eibe Frank. 2011. Classifier Chains for Multi-label Classification. *Machine Learning* 85, 3 (2011), 333–359.
[8] Stephanie Rude, Eva-Maria Gortner, and James Pennebaker. 2004. Language Use of Depressed and Depression-Vulnerable College Students. *Cognition and Emotion* 18, 8 (2004), 1121–1133.
[9] Vanessa Souza, Jéferson Nobre, and Karin Becker. 2021. A Deep Learning Ensemble to Classify Anxiety, Depression, and their Comorbidity from Texts of Social Networks. *J. Information and Data Management* 12, 3 (2021).
[10] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, et al. 2016. The FAIR Guiding Principles for Scientific Data Management and Stewardship. *Scientific Data* 3 (2016), 160018.